

Human motion analysis by measuring shape subspace variation

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Abstract

This paper proposes a framework for analyzing human motion using motion capture data. The core idea of our framework is to compactly represent a data set as a three-dimensional shape subspace and to geometrically measure the temporal variation of this shape subspace using its difference subspace. The difference subspace represents the difference between the two subspaces. Based on the proposed framework, we introduce three useful functions to measure 1) the temporal variation of the whole human motion, 2) the variation of each landmark, and 3) the coordination of movements between different parts of the human body. We validate the effectiveness of the proposed framework using the CMU Graphics Lab Motion Capture Database, which is a public dataset.

1 Introduction

Human motion capture data consist of a set of 3D landmarks representing joint points of a person in motion. Due to its ease of modeling, it has been widely used for various motion analyses, including sports [1],[2], rehabilitation [3],[4], and action recognition [5],[6],[7]. A set of p 3D landmarks can be compactly represented as a three-dimensional subspace, known as a shape subspace, in a p -dimensional vector space [8],[9],[10]. This shape subspace has the characteristic of being invariant to linear affine transformations. This means that the shape subspace for a given set of landmarks is uniquely determined regardless of the viewpoint from which they are observed; it does not depend on the 3D coordinate system. However, human motion involves nonlinear variations that cannot be captured by linear affine transformations, causing inevitable fluctuations of the shape subspace. The core idea of the proposed framework is to extract these motion-induced variations of the shape subspace as a difference subspace between two shape subspaces [11] as shown in Figure 1. Based on this framework, we propose the following three measurement functions:

Function 1: Measure how the whole shape of the human body changes over time depending on the type of motion.

Function 2: Measure the contribution of each landmark to the whole shape variation.

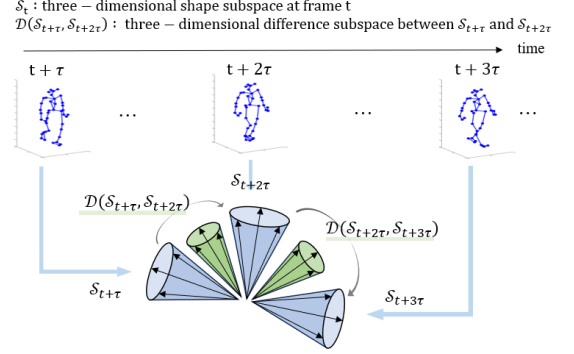


Figure 1: A set of p landmarks is represented by a three-dimensional shape subspace. The time variation of the whole motion from the $t + \tau$ th frame to the $t + 2\tau$ th frame is represented by the difference subspace $D(S_{t+\tau}, S_{t+2\tau})$ between the shape subspace $S_{t+\tau}$ and $S_{t+2\tau}$.

Function 3: Measure how different body parts, such as legs and arms, coordinate and vary together.

These functions aim to capture motion-specific periodicity and symmetry by analyzing geometric changes in the human body. They provide valuable insights for motion analysis. In particular, evaluating human gait in terms of periodicity and symmetry can effectively diagnose diseases associated with gait abnormalities [12],[13]. Traditional human motion analysis involves complex computations of landmark velocities, the angles between pairs of landmarks, and positional relationships. In contrast, the proposed method simplifies this process by leveraging the geometric structure of low-dimensional subspaces within a vector space. This simplicity allows us to address more intricate human motions without altering the core framework.

In this paper, we evaluate the effectiveness of each function using the CMU Graphics Lab Motion Capture Database [14] as a preliminary step in implementing such anomaly detection and motion classification.

2 Fundamental Techniques

2.1 Shape subspace of human body landmarks

Given a 3D object consisting of p landmarks, we can generate a shape subspace from them in the fol-

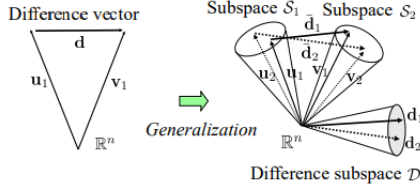


Figure 2: $\mathcal{D}$ represents the difference subspace between $\mathcal{S}_1$ and $\mathcal{S}_2$ [11].

lowing. We first set the $p \times 3$ matrix $\mathbf{X}_c$ defined as follows:[8],[9],[10].

$$\mathbf{X}_c = \mathbf{X} - \frac{1}{p} \sum_{i=1}^p \mathbf{r}_i, \quad (1)$$

$$\mathbf{X} = (\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_p)^T = \begin{pmatrix} x_1 & y_1 & z_1 \\ \vdots & \vdots & \vdots \\ x_p & y_p & z_p \end{pmatrix}, \quad (2)$$

where $\mathbf{r}_p = (x_p, y_p, z_p)^T$ is the p -th landmark on the object. Then, we can obtain the orthonormal basis $\mathbf{U} \in \mathbb{R}^{p \times 3}$ of the shape subspace, which we can calculate by applying Gram-Schmidt orthogonalization to the column vector of the matrix $\mathbf{X}_c$.

2.2 Difference subspace

2.2.1 Definition and Generation of Difference subspace

As shown in Figure 2, the difference subspace $\mathcal{D}(\mathcal{S}_1, \mathcal{S}_2)$ between two subspaces $\mathcal{S}_1$ and $\mathcal{S}_2$ is a generalization of the difference vector $\mathbf{d}$ between the unit vectors $\mathbf{u}$ and $\mathbf{v}$. It represents the difference component between the subspaces $\mathcal{S}_1$ and $\mathcal{S}_2$. Analytically, the difference subspace is defined as the remaining subspace of the sum space of $\mathcal{S}_1$ and $\mathcal{S}_2$, excluding the principal component subspace (common subspace)[11].

Given the d_1 -dimensional subspace $\mathcal{S}_1$ and the d_2 -dimensional subspace $\mathcal{S}_2$ in an p -dimensional vector space that have a r -dimensional intersection subspace ($d_1 \leq d_2$). The case $r = 0$ corresponds to the case where there is no overlap. The orthonormal basis of the difference subspace $\mathcal{D}(\mathcal{S}_1, \mathcal{S}_2)$ is independent of the existence of overlaps and can be obtained using singular value decomposition $\Phi^T \Psi = \mathbf{U} \Sigma \mathbf{V}^T$, where $\Phi \in \mathbb{R}^{n \times d_1}$, $\Psi \in \mathbb{R}^{n \times d_2}$ are the orthonormal bases of $\mathcal{S}_1, \mathcal{S}_2$ respectively, $\mathbf{U} \in \mathbb{R}^{d_1 \times d_1}$, $\Sigma \in \mathbb{R}^{d_1 \times d_2}$, $\mathbf{V} \in \mathbb{R}^{d_2 \times d_2}$, $\text{diag}(\Sigma) = \{1, \dots, 1, \cos \theta_1, \dots, \cos \theta_{d_1-r}\}$. Then, to target only non-zero canonical angles, we create $\mathbf{U}^* \in \mathbb{R}^{d_1 \times (d_1-r)}$, $\mathbf{V}^* \in \mathbb{R}^{d_2 \times (d_2-r)}$ by removing the column vectors of $\mathbf{U}$ and $\mathbf{V}$ corresponding to singular value 1 and $\text{diag}(\Sigma^*) = \{\cos \theta_1, \dots, \cos \theta_{d_1-r}\}$.

The orthonormal basis $\mathbf{D}$ of the difference subspace $\mathcal{D}$ is obtained using these matrices as follows.

$$\mathbf{D} = (\Phi \mathbf{U}^* - \Psi \mathbf{V}^*) \{2(\mathbf{I} - \Sigma^*)\}^{-\frac{1}{2}}, \quad (3)$$

where $(\Phi \mathbf{U}^* - \Psi \mathbf{V}^*)$ represents $[\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_{d_1-r}] \in \mathbb{R}^{n \times (d_1-r)}$ and $\{2(\mathbf{I} - \Sigma^*)\}^{-\frac{1}{2}}$ is $\text{diag}(\|\mathbf{d}_1\|_2, \dots, \|\mathbf{d}_{d_1-r}\|_2)^{-1} \in \mathbb{R}^{(d_1-r) \times (d_1-r)}$.

2.2.2 Magnitude of Difference subspace

The difference subspace is a linear subspace. Therefore, there is no mathematical definition for its magnitude, while the directions of its basis vectors have meaning. In this paper, corresponding to the magnitude of the difference vector $\|\mathbf{u} - \mathbf{v}\|_2^2$, the magnitude of the difference subspace $\mathcal{D}(\mathcal{S}_1, \mathcal{S}_2)$ of $\mathcal{S}_1$ and $\mathcal{S}_2$ is defined as follows[11].

$$\text{Mag}(\mathcal{D}(\mathcal{S}_1, \mathcal{S}_2)) = \text{Trace}(2(\mathbf{I} - \Sigma^*)) = \sum_{i=1}^{d_i} \bar{d}_i^2. \quad (4)$$

where $\bar{d}_i^2$ is the difference vector between the canonical vector pair $\mathbf{u}_i$ and $\mathbf{v}_i$ spanning the canonical angle θ_i .

3 Processing of each measurement function

3.1 Function 1: Measurement of whole Shape Variation

A set of the landmarks in frame t is represented as the shape subspace $\mathcal{S}_t$. The whole shape variation in the term from $t \sim t + \tau$ frames is obtained as $\text{Mag}(\mathcal{D}(\mathcal{S}_t, \mathcal{S}_{t+\tau}))$ using the magnitude defined by Equation (4). A larger magnitude indicates a greater shape variation.

3.2 Function 2: Contribution of Each Landmark to Motion Variability

We quantify the contribution of each of the p landmarks at frame t on the shape variation that occurs from frame t to frame $t + \tau$. The core idea of our calculation is to orthogonally project the shape subspace $\mathcal{S}_t$ onto the difference subspace $\mathcal{D}(\mathcal{S}_t, \mathcal{S}_{t+\tau})$. The shape subspace $\mathcal{S}_t$, which is spanned by the basis vectors $[\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3] \in \mathbb{R}^p$, is orthogonally projected onto the difference subspace $\mathcal{D}(\mathcal{S}_t, \mathcal{S}_{t+\tau})$ as $\mathbf{v}'_i = \mathbf{P} \mathbf{v}_i$, where $\mathbf{P}$ denotes the projection matrix onto the difference subspace $\mathcal{D}$.

Given the orthonormal basis vectors of $\mathcal{D}$ as $\mathbf{D} = [\mathbf{d}_1, \dots, \mathbf{d}_{d_1}]$, the projection matrix $\mathbf{P}$ is defined as $\mathbf{D} \mathbf{D}^T$. The vector $\mathbf{v}'_i$ projected onto the difference subspace represents the difference component [15]. Thus, the squared elements of each basis vector $[\mathbf{v}'_1, \mathbf{v}'_2, \mathbf{v}'_3]$ reflect their contribution to the whole motion. Accordingly, given $\mathbf{v}'_1 = [a_1, \dots, a_p]^T$, $\mathbf{v}'_2 = [b_1, \dots, b_p]^T$, $\mathbf{v}'_3 =$

$[c_1, \dots, c_p]^T$, we define the influence contribution _{i} of the i -th landmark p_i as $\|a_i\|^2 + \|b_i\|^2 + \|c_i\|^2$.

If the contribution of the i -th landmark is the largest, it indicates that this landmark has the greatest impact on the whole shape variation. By computing the influence of each landmark on the whole shape variation, we can identify body parts with significant variations during motion. For instance, in normal walking motion, it is expected that the landmarks corresponding to the legs and arms show large variations, whereas the head shows minimal variation.

3.3 Function 3: Analysis of coordination between body parts

The full-body landmarks are divided into multiple sub-parts, such as hands and feet, to measure the coordination of variations across different body parts. For example, we divide the landmarks of a full-body into two parts: right leg + upper body and left leg + upper body, targeting different walking styles (normal, intoxicated, and with an injured leg) as shown in Figure 5. Based on the structural similarity of the two generated shape subspaces, we measure how they change in temporal coordination.

Let $\mathcal{S}_t^r$ denote the shape subspace including the right leg at frame t , and $\mathcal{S}_t^l$ denote the shape subspace including the left leg. The structural similarity between these two shape subspaces is computed as follows:

$$\text{sim}(\mathcal{S}_t^r, \mathcal{S}_t^l) = \frac{1}{c} \sum_{i=1}^c \cos^2 \theta_i. \quad (5)$$

The orthonormal bases of $\mathcal{S}_t^r$ and $\mathcal{S}_t^l$ are given by $\mathbf{X} \in \mathbb{R}^{p \times 3}$ and $\mathbf{Y} \in \mathbb{R}^{p \times 3}$, respectively. The term $\cos \theta$ in the above equation corresponds to $\text{diag}(\mathbf{\Sigma}) = \{\cos \theta_1, \dots, \cos \theta_3\}$, obtained from the singular value decomposition $\mathbf{X}^T \mathbf{Y} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$.

For symmetrical movements such as normal walking, the structural similarity between the left and right shape subspaces $\mathcal{S}_t^r$ and $\mathcal{S}_t^l$ is expected to change periodically and regularly. This is because the human body is bilaterally symmetrical, and in normal walking, the left and right legs move alternately in sync. In contrast, an imbalance in the movement of the left and right legs disrupts this symmetry, leading to a breakdown in the periodic variation of structural similarity. The validity of this prediction will be examined in the experimental section.

4 Validity Verification of Functions

4.1 Experimental Conditions

In this experiment, we used the CMU Graphics Lab Motion Capture Database[14]. This large-scale dataset

contains various motion capture data, including everyday activities and sports movements. For the verification of Functions 1 and 2, we used three types of basic motion data: walking, running, and jumping. The number of landmarks for each data is 41, 330, and 41, respectively. To verify Function 3, we focused on three types of walking data: intoxicated walking, and walking with an injured leg. The corresponding number of landmarks for all cases is 41, respectively. As mentioned previously, the dimension of the shape subspace is set to 3, and τ is set to 5. This indicates that the magnitude of the difference subspace is calculated between two shape subspaces that are five frames apart.

4.2 Function 1: Measurement of whole Shape Variation

We visualized the whole shape variation for three basic motions: walking, running, and jumping, as shown in Figure 3. Figures 3a and 3b demonstrate that Function 1 effectively captures the periodic nature of both walking and running. The time-series variation reveals a significant increase in magnitude when the arms and legs are raised, which aligns with the expected skeletal structure and movement patterns of the human body.

In the case of jumping, we observe large variations at the beginning of the jump and upon landing, as illustrated in Figure 3c. Although these observations are qualitative, they are intuitively reasonable.

whole, Function 1 successfully employs shape subspaces and difference subspaces to accurately capture the variations in shape that correspond to different types of motion.

4.3 Function 2: Contribution of Each Landmark to Motion Variability

Figure 4 illustrates the contribution of each landmark to the whole shape. Points represented in warmer colors indicate a higher level of contribution. Throughout all movements, the head and chest show a minimal contribution, suggesting that variability in these areas is relatively low compared to others. In contrast, the contribution of the limbs exhibits higher contribution, demonstrating that their variability is more significant. Notably, the contribution of the limbs is especially pronounced during the arm and leg swings at the start and landing of a jump. These findings align with the dynamics of human motion, suggesting that Function 2 works effectively.

4.4 Function 3: Analysis of coordination between body parts

Using four walking datasets from the CMU Graphics Lab Motion Capture Database, including two types of NormalWalk, DrunkWalk, and HurtLegWalk, the landmarks were divided into two regions: right leg +

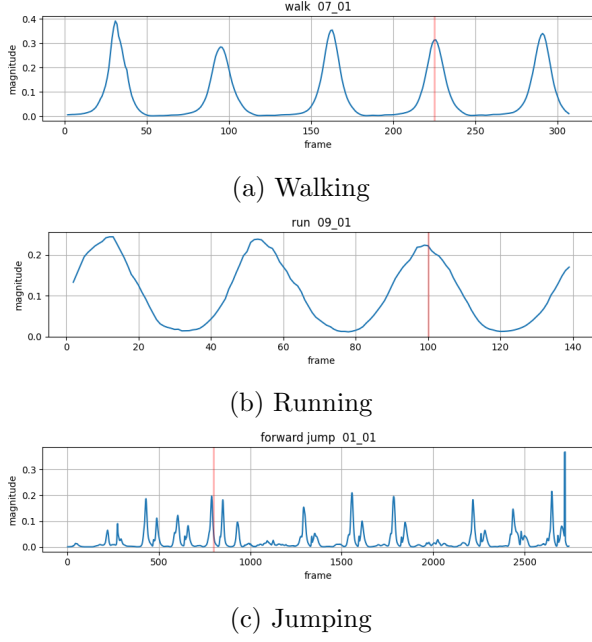


Figure 3: Variations in whole shape for each motion. The contribution of landmarks at the frames indicated by red lines is shown in Figure 4.

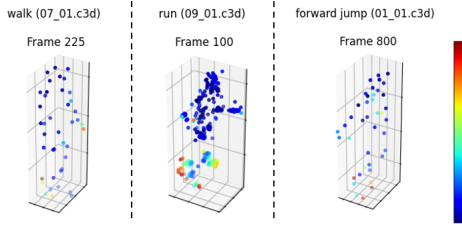


Figure 4: Influence of each landmark on the whole shape. (points closer to warm colors indicate higher influence).

upper body and left leg + upper body as shown in Figure 5. The temporal coordination between the two generated shape subspaces $\mathcal{S}^r$ and $\mathcal{S}^l$ was evaluated using the similarity metric defined by Equation (5). For the two types of NormalWalk, periodic variations similar to sine waves were observed (Figure 6). This is an expected result, as NormalWalk involves alternating leg movements, leading to consistent and symmetrical shape variation patterns on both sides of the body. In contrast, for HurtLegWalk and DrunkWalk, where symmetry is disrupted, an imbalance in the movements of the left and right legs causes the periodicity to become irregular. **These results confirm that Function 3 can extract differences in temporal coordination and periodicity between regions according to walking styles and show the possibility of medical applications such**

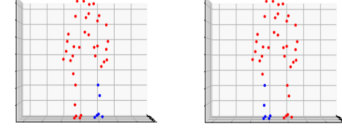


Figure 5: Left: Landmarks of (right leg + upper body), Right: Landmarks of (left leg + upper body). The red landmarks are used to create shape subspaces.

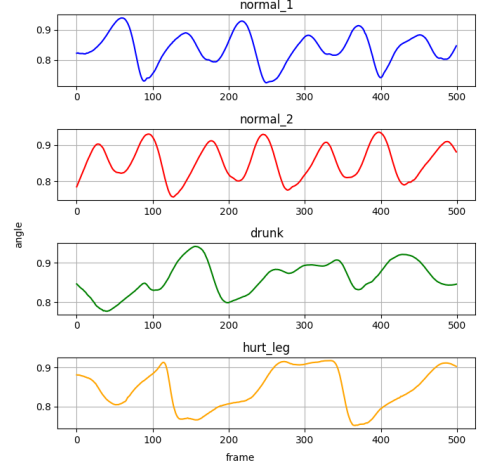


Figure 6: Similarity between the shape subspace $\mathcal{S}^r$ generated from the (right leg + upper body) landmarks and the shape subspace $\mathcal{S}^l$ generated from the (left leg + upper body) landmarks for each frame.

as the diagnosis of gait abnormalities in terms of periodicity and symmetry.

5 Conclusion

In this paper, we proposed a framework for precisely analyzing human motion from motion capture, representing a set of landmarks as a shape subspace. Then, we measured its temporal variations geometrically using differential subspaces. Based on this framework, we proposed three measurement functions: 1) Variation of whole body shape, 2) Contribution of each landmark to motion variability and 3) Coordination analysis of different sub-parts of the whole human body. **We validated the effectiveness of these functions and show the possibility of medical and other applications in simplified motion analysis.** In future work, we aim to perform quantitative evaluations of the proposed framework and develop a more practical system capable of real-time processing, from landmarks detection to visualization of variation analysis results.

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